What Are We Measuring? Benchmark Integrity and Zone-Based Reframing for Emotion Recognition in Autistic Children

Nikhil Mahesh
Pioneer Research

***Abstract*—Automated emotion recognition is widely proposed as an assistive technology for autistic children, and recent papers report up to 97.95% accuracy on public autism facial-expression benchmarks. Independently collected clinical corpora report 40–66% on the same task. We audit the public archive behind many of the former figures. Re-acquiring it, we find it ships two overlapping dataset trees sharing 45% of the smaller tree's images, that 21.4% of its files are byte-identical duplicates, and that 119 duplicate groups carry contradictory emotion labels on pixel-identical images. Holding features, classifier, and images fixed and varying only the split protocol, five-category balanced accuracy spans 0.346–0.750 across five defensible readings of the same archive. Duplicate leakage is not the main driver: a nearest-neighbour reference memoriser gains only 2.1 points from a split where 80.6% of test images have a family member in training, and reaches 0.755 at best. Exact-duplicate and augmentation-family leakage under these features therefore does not reach the published band. The label contradictions are sharply asymmetric. A permutation null preserving zone marginals expects 536 conflicted files, 217 of them touching positive/neutral; the archive has 280, of which *one* touches positive/neutral. Cross-valence conflict is depleted roughly 200-fold while within-negative conflict runs near chance: one boundary, valence, absorbs 102 of 103 contradictions. Controls then refute our own prior framework—the three-zone partition absorbs none of them, because it cuts between anger and sadness, and its apparent protocol-stability gain is generic to class count rather than to those classes. The granularity this archive supports is binary valence; anything finer needs context beyond the face. This is a measurement audit, not a deployable classifier, and not a claim about any child's internal state.**

***Index Terms*—benchmark integrity, data leakage, label noise, affective computing, autism**

I. Introduction

Roughly 1 in 31 eight-year-olds in the United States is diagnosed with autism spectrum disorder (ASD) [1]. Because emotional expression in autistic people is frequently misread by non-autistic observers—the *double empathy problem* [2]—automated emotion recognition is repeatedly proposed as a communication aid.

The literature on this task is in an unusual state. Papers evaluating on public autism facial-expression archives report accuracies from 80.0% [3] to 97.95% [4]. Papers evaluating on independently collected, consented clinical corpora report far less: 40% for a commercial analyser on the EMBOA autism corpus [5], 57.95–64.92% for four commercial classifiers on parent-reported autistic children [6], and 66.43% for a classifier trained on typically developing children and tested on autistic children [7]. The gap tracks how the evaluation data were obtained rather than what the models are.

That correlation has become hard to ignore. Publishers have retracted work built on scraped autism face datasets, including the source paper for the archive we audit [8] and a facial emotion recognition paper [9], on the grounds that images were collected from the internet without documented diagnosis, ethical oversight, or guardian consent. That established a *provenance* failure. It did not establish whether the reported numbers were also measurement artifacts.

This paper supplies that measurement. Our contributions are:

1) A quantified integrity audit of a widely used public autism emotion archive, computed from the vendor's own directory layout (Sec. III).
2) A protocol ablation holding features, classifier, and images fixed while varying only the split, plus a nearest-neighbour reference memoriser that probes how much the duplicates are worth (Sec. IV).
3) A partition analysis showing that one boundary—valence—absorbs 102 of 103 label contradictions, supported by a permutation null (Sec. V).
4) A control experiment that refutes our own protocol-stability argument for three-zone labels, and the narrower granularity claim that survives it (Sec. VI).

We audit an instrument rather than build one. We make no claim about any child's internal state, and we treat every label as a filing decision about an image rather than as ground truth about a person.

II. Related Work

Cross-population transfer is established. Grossard *et al.* [7] ran the controlled experiment on lab-collected video from 157 typically developing and 36 autistic children with subject-independent cross-validation, obtaining 82.05% within-population against 66.43% across. Gaya-Morey *et* *al.* [10] repeated the design with twelve CNNs on an adjacent population. We do not claim this finding; we ask whether the public benchmarks now used to measure it can support the measurement.

Benchmark contamination is a mature field. Kapoor and Narayanan [11] catalogue eight leakage types across 294 papers; Barz and Denzler [12] found 3.3% and 10% duplicate rates in CIFAR test sets; Northcutt *et al.* [13] showed label errors reorder rankings, and Recht *et al.* [14] that the evaluation set alone moves accuracy. We adopt the taxonomy of Ramos *et al.* [15]—*hard* leakage for identical images across a split, *soft* for near duplicates, *intra-dataset* for overlap within one dataset's splits—and the notions of *augmented duplicates* [16] and *synthesis leakage* [17], where train and test items derive from a common source under a transform. None of this work examines a clinical affect dataset.

Contamination in this archive has been asserted, not measured. The authors of the FERAC derivative [18] reported "class overlap and image duplication" here, removed duplicates, and dropped two classes as unrecoverable. They reported no detection method, threshold, duplicate rate, leakage rate, or effect of cleaning. Two groups have now flagged the same archive; what remains open is the quantity.

Dataset critique. Birhane and Prabhu [19] argue scraped corpora containing people, including minors, are ethically defective at collection; Gebru *et al.* [20] and Paullada *et al.* [21] propose the documentation practices absent here.

Why the labels might be the problem. Barrett *et al.* [22] challenge the premise that emotional state is readable from facial configuration at all. Keating *et al.* [23], from over 265 million motion-capture points, find autistic and non-autistic expressions differ in which muscles carry each emotion. If expressions are ambiguous under a non-autistic reading, forcing them into mutually exclusive categories should produce contradictions concentrated where categories are closest.

III. The Archive

We audit the public Kaggle archive associated with [8].[1] The 1,672 SHA-256 digests recorded in an earlier run of ours are all present in the fresh download, confirming the archive has not changed under us; that run excluded `surprise` and the conflicted groups, so it covered a subset of the 1,774 distinct digests audited here. Statistics below come from the vendor's own directory layout, before any re-splitting: this is what a downstream user receives.

Two overlapping datasets ship as one. The archive contains two top-level trees (Table I) with different class balances and different splits, sharing 323 unique images—45.0% of the smaller tree. This plausibly explains a puzzling feature of the literature: published descriptions of this dataset report its size variously as 348, 830, 1,603, and "2500+" images, consistent with different papers loading different trees.

TABLE I
Two dataset trees shipped in one archive.

Tree	Files	Unique	Train	Test
Balanced copy	1,425	1,380	1,199	226
Imbalanced copy	833	717	758	75
Shared unique images: 323 (45.0% of smaller tree)				

One file in five is a duplicate. Of 2,258 labelled images only 1,774 are distinct by SHA-256; 484 files (21.4%) are byte-identical copies, in groups reaching size 5. Perceptual hashing at Hamming distance 4 adds 163 near-duplicate components. A filename suffix (`_aug_N`) identifies 1,020 augmentation families, 335 larger than one, the largest containing 9, so a substantial part of the archive is synthetic variation on fewer source images—synthesis leakage in the sense of [17]. Notably these families are identifiable from filenames alone, so the duplication was detectable without any image analysis.

Pixel-identical images carry contradictory labels. Under the vendor's six categories, 119 byte-identical groups (327 files) contain members filed under more than one emotion. Establishing this requires no external ground truth and no annotator panel, only a hash: it is an internal inconsistency of the artifact, which is a cheaper and stronger claim than label-error detection in the sense of [13].

IV. How Much of the Number Is the Protocol?

Contamination matters only if it moves results. We hold the images, the features, and the classifier fixed and vary *only* how train and test are separated. Features are frozen `ViT-B/16` embeddings [24]; the head is a one-hidden-layer MLP with epoch-wise early stopping; every protocol runs over ten seeds [25]. Because `surprise` has no zone, it is dropped from *both* arms, so the categorical arm is a five-way task over exactly the images the zone arm sees and the two columns differ only in label granularity.

We add a 1-nearest-neighbour classifier as a *reference memoriser*. It stores the training set exactly, so a test image whose byte-identical twin is in training is recovered with certainty. It is not an upper bound over all models—a fine-tuned network memorises and generalises, and we did not run one—but it is the estimator most able to convert an exact duplicate into a correct prediction, which makes it the sharpest available probe of how much these duplicates are worth under this representation.

Table II and Fig. 1 support three claims and refute a fourth we expected to make.

The reported number is largely a choice. Across five defensible protocols on identical images, five-category balanced accuracy spans 0.346 to 0.750—a 40.4-point range attributable entirely to how the archive is read. Seed dispersion is 2.3–6.3 points, an order of magnitude smaller than the protocol spread

[1] Slug `fatmamtalaat/autistic-children-emotions-dr-fatma-m-talaat`. We name it for reproducibility while noting it distributes images of minors of disputed provenance; readers should weigh that before downloading.

TABLE II

Same image pool, features, and classifier; only the split changes—so the test sets differ in size and composition, and n is given for each. "Grp" is the share of test images whose duplicate or augmentation family also appears in training. Ten seeds; sd is over seeds and does not include test-set sampling error, which at $n = 69$ is roughly ± 0.06.

Protocol	Bal. acc.	sd	1-NN	Grp	n
Five categories (chance $= 0.200$)					
Vendor split, imbal. tree	0.346	.050	0.348	26.1%	69
Train one tree, test other	0.500	.043	0.476	24.6%	69
Vendor split, bal. tree	0.558	.037	0.585	33.3%	189
Group-aware	0.700	.052	0.734	0.0%	180
Pooled random	0.750	.028	0.755	80.6%	293
Three zones (chance $= 0.333$)					
Vendor split, imbal. tree	0.497	.063	0.392	26.1%	69
Vendor split, bal. tree	0.562	.025	0.552	33.3%	189
Train one tree, test other	0.665	.025	0.661	24.6%	69
Group-aware	0.727	.034	0.733	0.0%	180
Pooled random	0.735	.023	0.736	80.6%	293

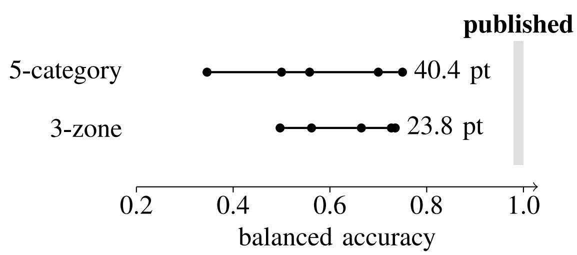

Fig. 1. Every dot is the same images, features, and classifier under a different split protocol. The spread is the protocol, not the model. No protocol—including a 1-NN memoriser on the most contaminated split—comes within 22 points of the published band.

even after allowing that the range of five draws is itself about 2.3σ. A reader given one number cannot tell where in that band it sits.

Zones are markedly more protocol-stable. The same protocols span 23.8 points under the zone collapse against 40.4 under five categories. Because the two label spaces have different chance floors, we compare them chance-corrected as $(\mathrm{BA}-1/k)/(1-1/k)$: the ranges are 0.357 and 0.505, a **29.3%** reduction in the protocol-induced range. The coarser target is not merely easier; it is less sensitive to a methodological choice the reader cannot see.

Duplicate leakage is real but second-order. Pooled random leaks the family of 80.6% of its test images and exceeds the group-aware split by 5.0 points under five categories and 0.8 under zones. The reference memoriser gains even less: 1-NN improves by only 2.1 and 0.3 points respectively from the same contamination, so the duplicates are worth little even to the estimator best placed to exploit them.

TABLE III

Byte-identical duplicate groups still carrying conflicting labels after coarsening, against 346 duplicate groups and a 103-conflict five-way baseline.

Coarsening	Conflicts	Absorbed
Five categories (baseline)	103	—
{natural, joy} vs rest **(valence)**	**1**	**102**
{anger, sadness} vs rest	26	77
{anger} vs rest	77	26
{anger, fear} vs rest	102	1
Valence zones {nat,joy}{ang,fear}{sad}	103	0
Best of 15 three-block partitions	26	77

This form of contamination does not reach the published band. The highest value any protocol reaches, under any estimator including the reference memoriser, is 0.755—below the 80.0–97.95% band reported on archives of this family [3], [4], and 22 points below its upper end. So *exact-duplicate and augmentation-family leakage, under frozen* `ViT-B/16` *features and our re-splits*, does not by itself produce those figures. We are careful not to over-read this. It is not a claim that contamination is irrelevant: augmentation applied before splitting in downstream derivatives is itself synthesis leakage [17], and it would not appear in our accounting because we audit the archive as shipped rather than the augmented derivatives built from it. Small test sets and selection among runs are further candidates. A fine-tuned network on a contaminated split remains untested and is the clearest target for follow-up.

V. Where the Contradictions Live

The contradictions are not spread evenly across the label space, and locating them turns out to matter more than counting them.

Restrict to the 1,952 images whose category maps to a valence zone (`surprise` has none) and ask, for each way of coarsening those five categories, how many byte-identical duplicate groups still straddle a block boundary. A coarsening that absorbs many contradictions is one whose boundaries the filing process respected. Because every candidate below has the same block shape, this measures *where* the boundary falls, not how many classes there are. Table III gives the result.

One boundary absorbs almost everything. Splitting positive/neutral from negative leaves one conflicted group out of 346; every other binary cut leaves far more. A permutation null preserving the zone marginals and duplicate-group structure sharpens this: it expects 536 conflicted files, 217 touching positive/neutral, where the archive has 280 of which exactly one does. Cross-valence conflict is depleted roughly 200-fold, while conflict *within* negative affect sits at 88% of its null expectation—chance. The filing process separated valence almost perfectly and anger, fear, and sadness not at all.

The three-zone partition absorbs none of it. This is a negative result about our own prior framework. Valence zones place anger and fear together and sadness apart, but anger versus sadness is the most common contradiction in the archive, so the zone-2/zone-3 boundary runs straight through the region of maximum disagreement. Among the 15 same-shape partitions the zone partition ties for worst at absorbing contradictions, while grouping anger with sadness absorbs 77 of 103.

We do not describe this as annotator disagreement. The archive is a scrape with no documented labelling procedure, and a simpler mechanism is available: one image retrieved under two queries and written into two folders. Either way the consequence for a model is the same, and it is the consequence, not the mechanism, our evidence supports.

VI. What This Does and Does Not License

A stability argument for zones does not survive its control. Collapsing five classes to three narrows the chance-corrected protocol range from 0.482 to 0.404, a 16.1% reduction. That reads as support for zones until the control is run. Across all 15 partitions of the same 2-2-1 shape the chance-corrected range runs from 0.325 to 0.628; the zone partition ranks 5th, and a deliberately valence-crossing partition does *better* (0.391, an 18.7% reduction). Fewer classes buys stability; *these* classes buy nothing extra. We report this because we expected the opposite, and because a coarsening cannot be justified by a property that every coarsening shares.

What survives is a granularity claim, and it is sharper. The evidence in Sec. V is specific to one boundary: on this archive the reproducible distinction is valence, and the anger/fear/sadness distinctions are not reproducible at all. That argues for a *binary* valence target rather than the three-zone scheme we and others have used—or, if three levels are wanted for intervention purposes, for a partition that does not cut between anger and sadness. The prior framework had the direction right and the partition wrong, and only the audit could show which.

We stop short of the actionability claim. Zones are usually motivated by what a caregiver would do—de-escalate, engage, comfort—and we ran no caregiver study; nothing here shows that anger and fear share a response. What we can say is which distinctions the data support.

Context beyond the face. Even at its cleanest this task saturates well below deployment quality: the group-aware five-category result is 0.700, and Barrett *et al.* [22] argue facial configuration underdetermines emotional state in principle. If the face reliably yields only valence, then everything clinically interesting—which negative state, and why—must come from elsewhere.

This is where multimodal and vision-language approaches earn their place, on a data argument rather than an accuracy one. A face-only autism-specific model needs autism-specific face data—precisely the resource whose acquisition produced the retractions cited above. A vision-language model brings prior knowledge of scenes, activities, and posture without any autism-specific corpus, and supplies the context a cropped face cannot. Reporting a coarse state with an explanation a caregiver can check is also more auditable than a fine-grained category with a confidence score, when those categories are not reproducible.

VII. Limitations and Ethics

This audits one archive and does not establish prevalence, though [18] reports consistent qualitative findings. Duplicate detection is hash-based and will miss re-encoded or cropped duplicates, so our rates are lower bounds; the Hamming-4 threshold was not validated against hand-labelled pairs. The archive supplies no subject identities, so our groups are duplicate and augmentation families, not people, and a subject-disjoint split would be stricter. Our features come from a web-pretrained backbone, so we cannot exclude that the encoder has seen these images—a caveat applying to every published result here, which our "0% leakage" rows do not escape. The reference memoriser characterises contamination under *these* features only; it is not an upper bound over models, and no fine-tuned network was run.

Ethics. This is secondary analysis of an already-public dataset, with no interaction with human subjects and no attempt to re-identify anyone. We did not seek an institutional determination and do not assert one here. We report only aggregate statistics: no image, filename, hash, or per-image prediction is released. The image corpus was deleted after hashing and feature extraction. We make no claim that any depicted child is autistic, that any label reflects a felt emotion, or that any system should be deployed on this basis; where Sec. V references findings about autistic expression, it does so to motivate a hypothesis, not to assert that this archive depicts diagnosed children. The archive's provenance is disputed and its license unstated, so we redistribute nothing.

VIII. Conclusion

A widely used public autism emotion archive ships two overlapping copies sharing 45% of the smaller, is 21.4% duplicated, and contains 103 groups of pixel-identical images with contradictory labels. Varying only the split protocol moves five-category balanced accuracy across a 40.4-point band, so a single number on this archive says little about a model. Exact-duplicate and augmentation-family leakage is not the main driver—worth 5.0 points, and 2.1 to a reference memoriser—and no protocol we ran reaches the published band, though augmentation applied before splitting in downstream derivatives would not show up in our accounting.

One boundary carries the label structure: valence absorbs 102 of 103 contradictions, and cross-valence conflict is depleted roughly 200-fold against a marginal-preserving null. Our own three-zone framework absorbs none of them, because it cuts between anger and sadness where disagreement is greatest, and its protocol-stability advantage disappears against same-shape controls. The granularity this archive supports is binary valence; anything finer needs context beyond the face. Any future benchmark here should report a duplicate audit, a group-aware split, and chance-corrected metrics as a condition of being believed—and, given how this archive was assembled, should not be built by scraping.

References

[1] K. A. Shaw *et al.*, "Prevalence and early identification of autism spectrum disorder among children aged 4 and 8 years," MMWR Surveillance Summaries, vol. 74, no. 2, Centers for Disease Control and Prevention, 2025.

[2] National Autistic Society, "The double empathy problem," https://www.autism.org.uk/advice-and-guidance/professional-practice/double-empathy, 2018.

[3] P. Radočaj and G. Martinović, "Emotion recognition in autistic children through facial expressions using advanced deep learning architectures," *Applied Sciences*, vol. 15, no. 17, p. 9555, 2025.

[4] S. Hanumantharayappa and M. R. Bharamagoudra, "Confidence-weighted ensemble learning and graph-based shapley theory for refined emotion prediction in autistic children," *Journal of Trends in Computer Science and Smart Technology*, vol. 8, no. 3, pp. 745–767, 2026.

[5] D. Kiejdo, M. Depka Prądzinska, and T. Zawadzka, "Late fusion model for emotion recognition from facial expressions and biosignals in a dataset of children with autism spectrum disorder," *Sensors*, vol. 25, no. 24, p. 7485, 2025.

[6] H. Kalantarian, K. Jedoui, K. Dunlap, J. Schwartz, P. Washington, A. Husic, Q. Tariq, M. Ning, A. Kline, and D. P. Wall, "The performance of emotion classifiers for children with parent-reported autism: Quantitative feasibility study," *JMIR Mental Health*, vol. 7, no. 4, p. e13174, 2020.

[7] C. Grossard, A. Dapogny, D. Cohen, S. Bernheim, E. Juillet, F. Hamel, S. Hun, J. Bourgeois, H. Pellerin, S. Serret, K. Bailly, and L. Chaby, "Children with autism spectrum disorder produce more ambiguous and less socially meaningful facial expressions: an experimental study using random forest classifiers," *Molecular Autism*, vol. 11, p. 5, 2020.

[8] F. M. Talaat, "Retracted article: Real-time facial emotion recognition model based on kernel autoencoder and convolutional neural network for autism children," *Soft Computing*, 2023, retracted 16 November 2025.

[9] İ. Güngör, A. Sarman, and S. Tuncay, "Retracted: Can artificial intelligence and face recognition using deep learning detect emotions in children with autism?" *PLOS ONE*, 2025, retracted 23 December 2025.

[10] F. X. Gaya-Morey, S. Ramis, J. M. Buades-Rubio, and C. Manresa-Yee, "Assessing the efficacy of deep learning approaches for facial expression recognition in individuals with intellectual disabilities," *Multimedia Systems*, 2026.

[11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[12] B. Barz and J. Denzler, "Do we train on test data? purging CIFAR of near-duplicates," *Journal of Imaging*, vol. 6, no. 6, p. 41, 2020.

[13] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.

[14] B. Recht, R. Roelofs, L. Schmidt, and V. Shankar, "Do ImageNet classifiers generalize to ImageNet?" in *Proc. Int. Conf. Machine Learning (ICML)*, 2019.

[15] P. Ramos, R. Ramos, and N. Garcia, "Data leakage in visual datasets," in *Proc. IEEE/CVF Int. Conf. Computer Vision Workshops (ICCVW)*, 2025, arXiv:2508.17416.

[16] Y. K. Adimoolam, C. Poullis, and M. Averkiou, "Data leakage detection and de-duplication in large scale geospatial image datasets," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2026.

[17] A. Apicella, F. Isgrò, and R. Prevete, "Don't push the button! exploring data leakage risks in machine learning and transfer learning," *Artificial Intelligence Review*, 2025.

[18] R. Chaiti *et al.*, "Emotion recognition of autistic children from facial images using hybrid model," in *Proc. IEEE Int. Conf.*, 2024.

[19] A. Birhane and V. U. Prabhu, "Large image datasets: A pyrrhic win for computer vision?" in *Proc. IEEE Winter Conf. Applications of Computer Vision (WACV)*, 2021.

[20] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[21] A. Paullada, I. D. Raji, E. M. Bender, E. Denton, and A. Hanna, "Data and its (dis)contents: A survey of dataset development and use in machine learning research," *Patterns*, vol. 2, no. 11, p. 100336, 2021.

[22] L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, and S. D. Pollak, "Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements," *Psychological Science in the Public Interest*, vol. 20, no. 1, pp. 1–68, 2019.

[23] C. T. Keating, S. Sowden-Carvalho, H. O'Donoghue, and J. L. Cook, "Mismatching expressions: Spatiotemporal and kinematic differences in autistic and non-autistic facial expressions," *Autism Research*, vol. 19, no. 2, p. e70157, 2026.

[24] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[25] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk *et al.*, "Accounting for variance in machine learning benchmarks," in *Proc. Machine Learning and Systems (MLSys)*, 2021.